

SHETH L.U.J. AND SIR M.V. DEGREE COLLEGE OF SCIENCE

SUBJECT NAME: Power BI And Tableau Practical

PRACTICAL - 4

Aim: Performing Data Analysis using Calculations in Tableau.

- a) Create basic calculated fields using formulas
- b) Perform row-level calculations
- c) Perform aggregate-level calculations
- d) Implement FIXED Level of Detail (LOD) calculations
- e) Use INCLUDE and EXCLUDE LOD expressions
- f) Create and use parameters for dynamic analysis
- g) Apply calculations to fix data issues and enhance insights

Dataset: Indian Store Data

A: Creating Basic Calculated Fields Using Formulas

1. Connecting to the Dataset

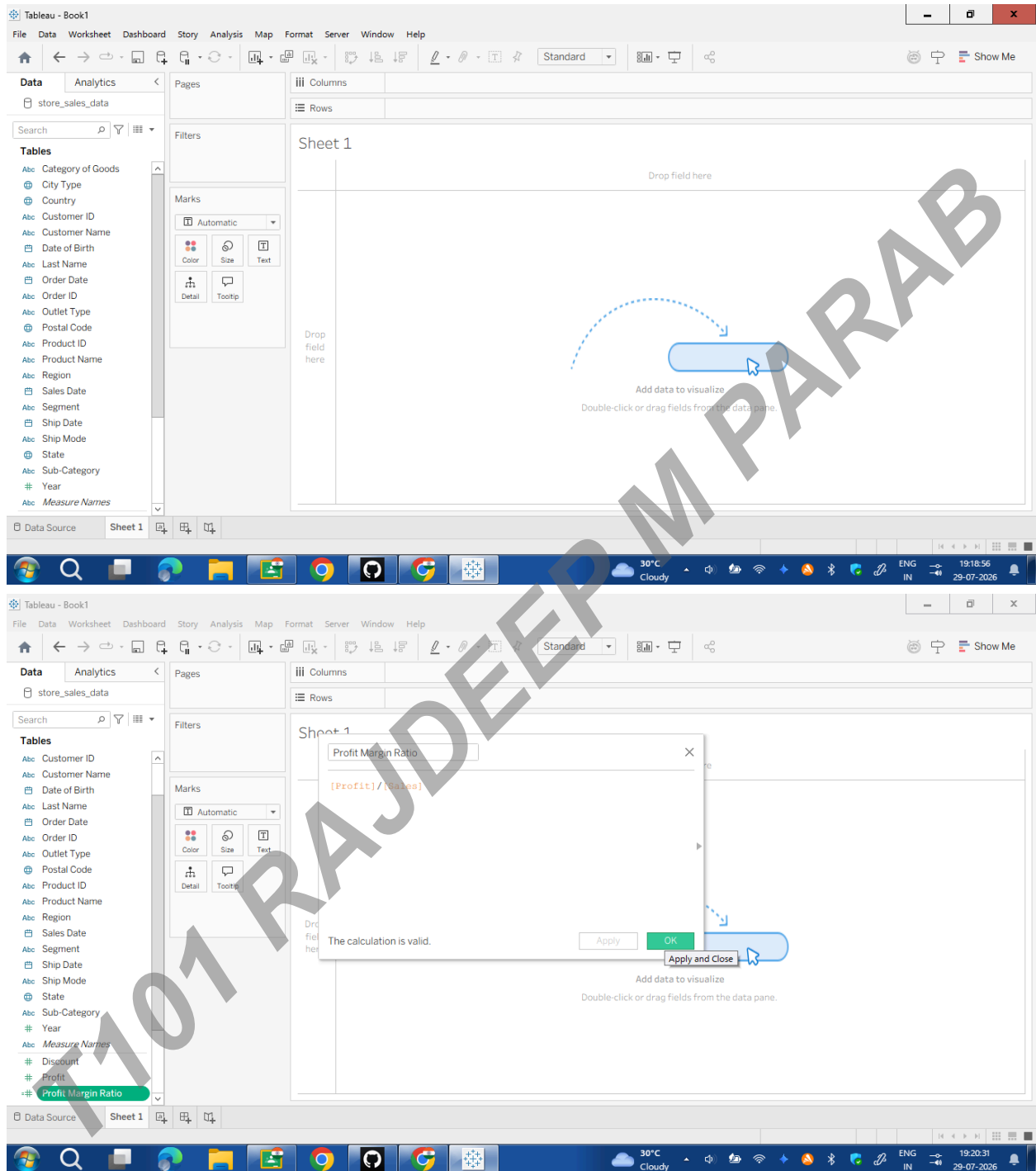
1. Launch Tableau. Under the Connect panel, click on Text File, select your store_sales_data.csv, and click Open.
2. Click on Sheet 1 at the bottom-left corner to enter your workspace.

2. Building a Basic Custom Calculation

1. Look at your left-hand Data Pane. Right-click anywhere in the blank space of the pane (or click the small downward triangle next to the search bar) and select Create Calculated Field...
2. In the calculation editor window that pops up, clear any default text and name the field: Profit Margin Ratio.
3. In the main workspace box, enter the following basic mathematical formula:
4. Look at the bottom-left corner of the editor box. It should say "The calculation is valid." Click OK.
5. Notice that your new custom field Profit Margin Ratio now appears in your Data Pane list with a small equal sign (=) next to its data type icon, showing it is a calculated field.

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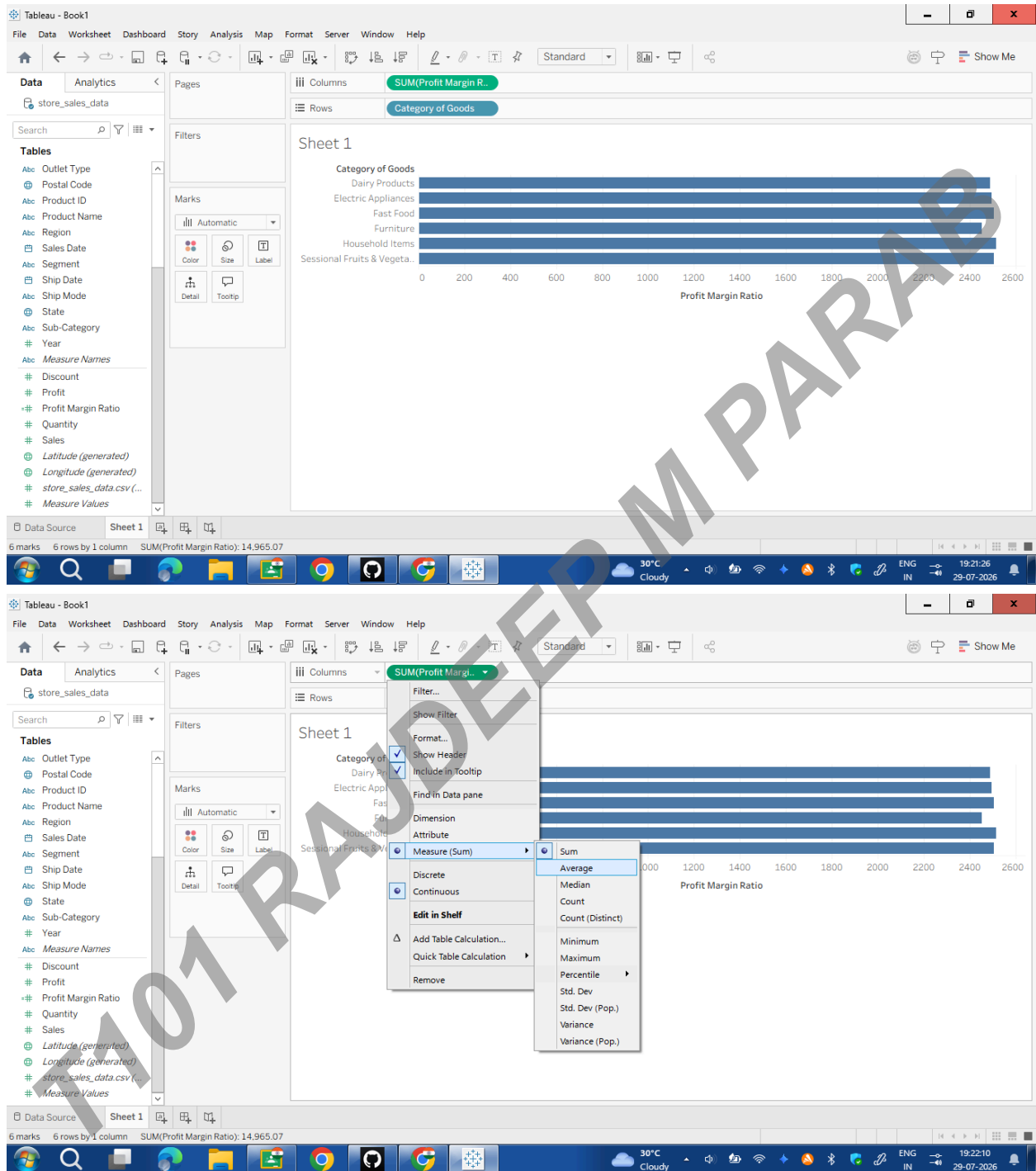
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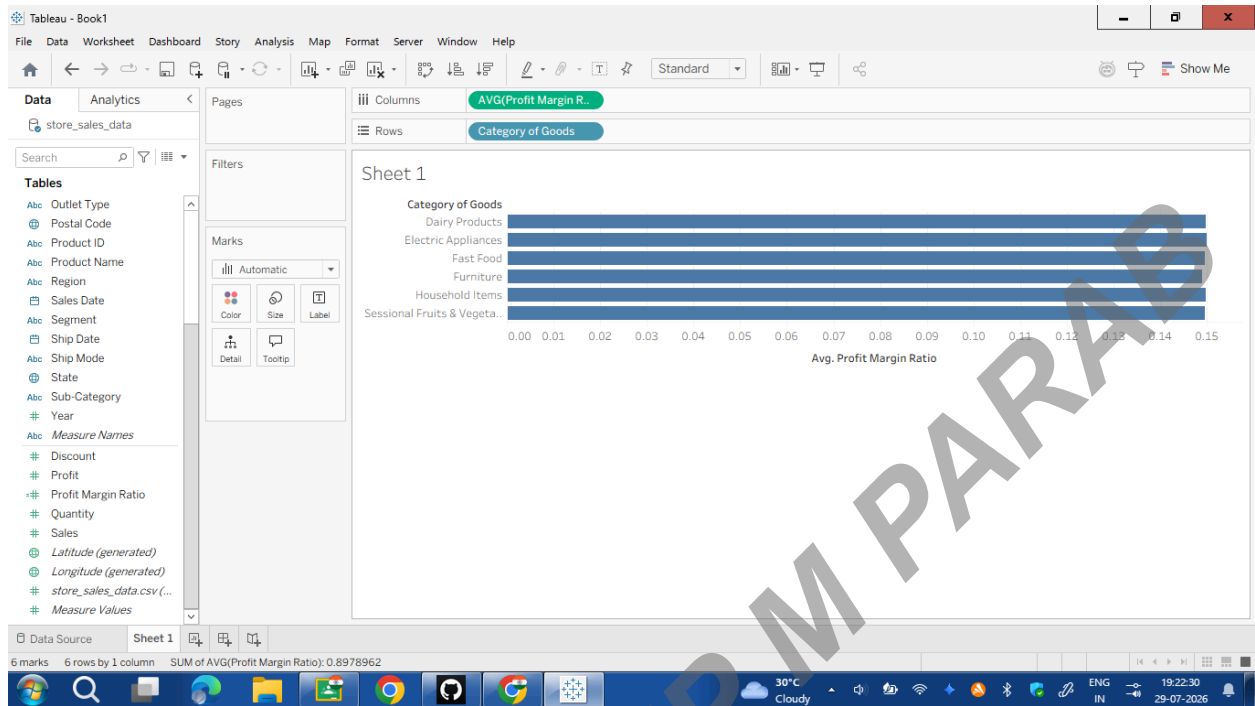
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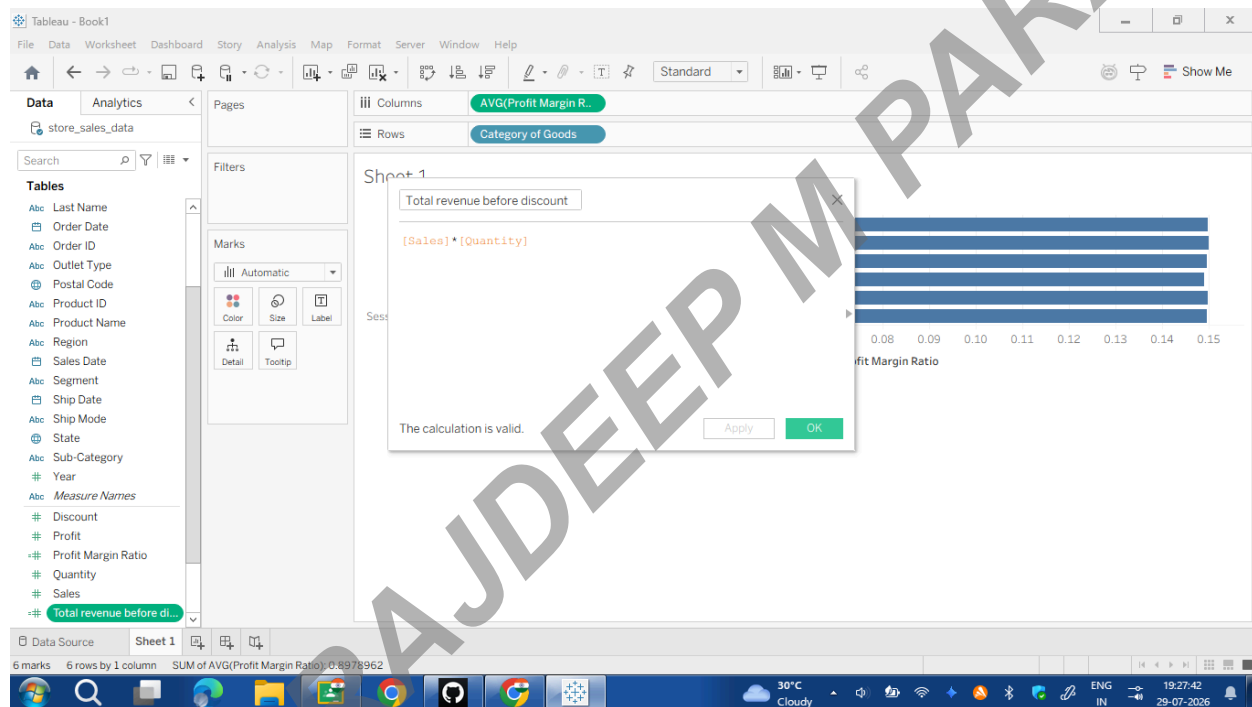
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B: Performing Row-Level Calculations

1. Creating a Row-Level Calculated Field

1. Right-click anywhere in the blank white space of your Data Pane and choose Create Calculated Field...
2. Name the field: Total Revenue Before Discount.
3. In the calculation formula canvas area, input the following expression: $[Sales] * [Quantity]$
4. Verify the message at the bottom reads "The calculation is valid" and click OK.

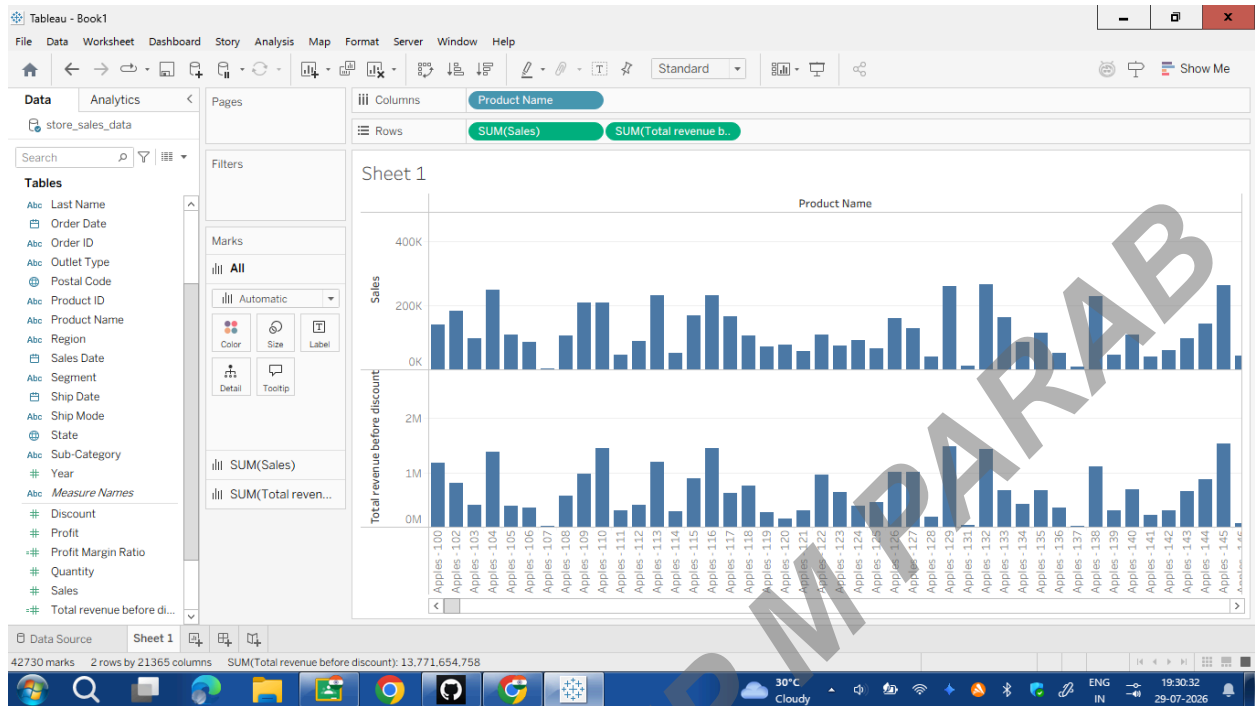


2. Putting the Field to Use

1. Drag the Product Name dimension from the Data Pane and drop it onto the Rows shelf.
2. Drag your newly created Total Revenue Before Discount calculated field and drop it onto the Columns shelf.
3. Drag the raw Sales measure pill and drop it directly onto the Columns shelf right next to the previous calculated pill to visually compare the two numbers.

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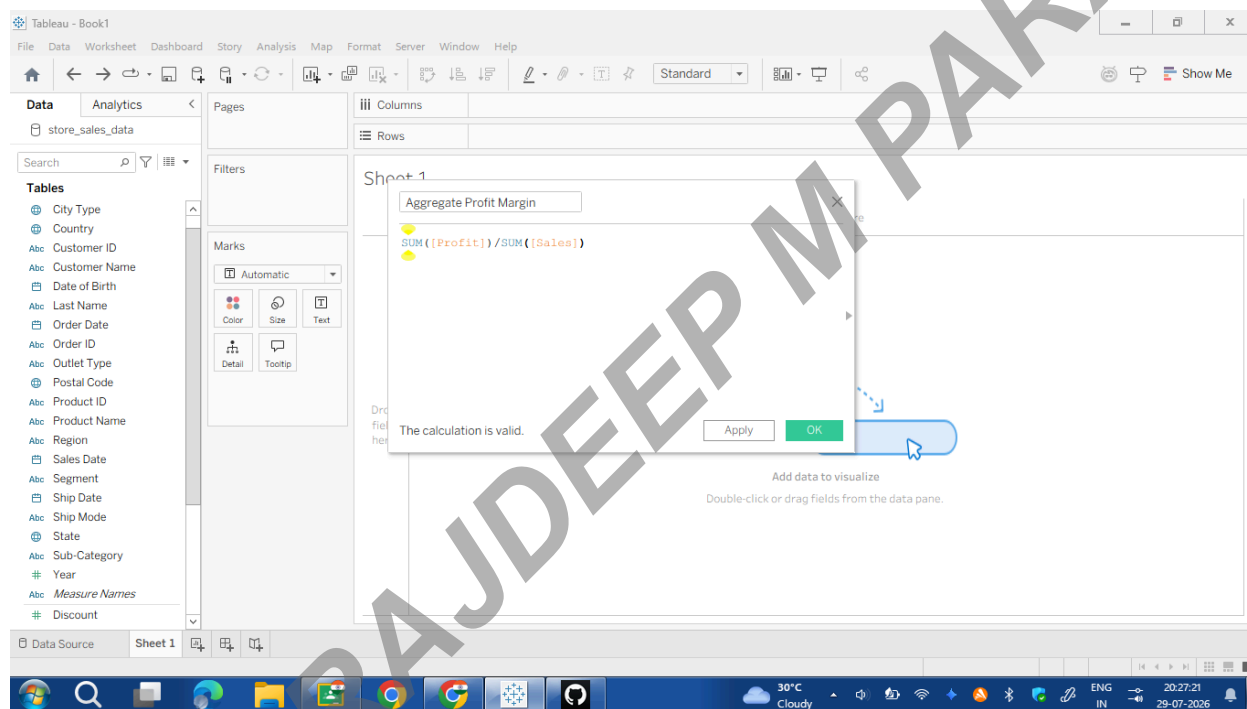
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C: Performing Aggregate-Level Calculations

1. Creating an Aggregate Calculated Field

1. Right-click in the blank white space of your Data Pane or click the Small Triangle Icon next to the search bar and select Create Calculated Field...
2. Name the field: Aggregate Profit Margin.
3. In the calculation formula window, type the following expression:
 $\text{SUM}([\text{Profit}]) / \text{SUM}([\text{Sales}])$
4. Look at the bottom left to ensure it reads "The calculation is valid" and click OK.

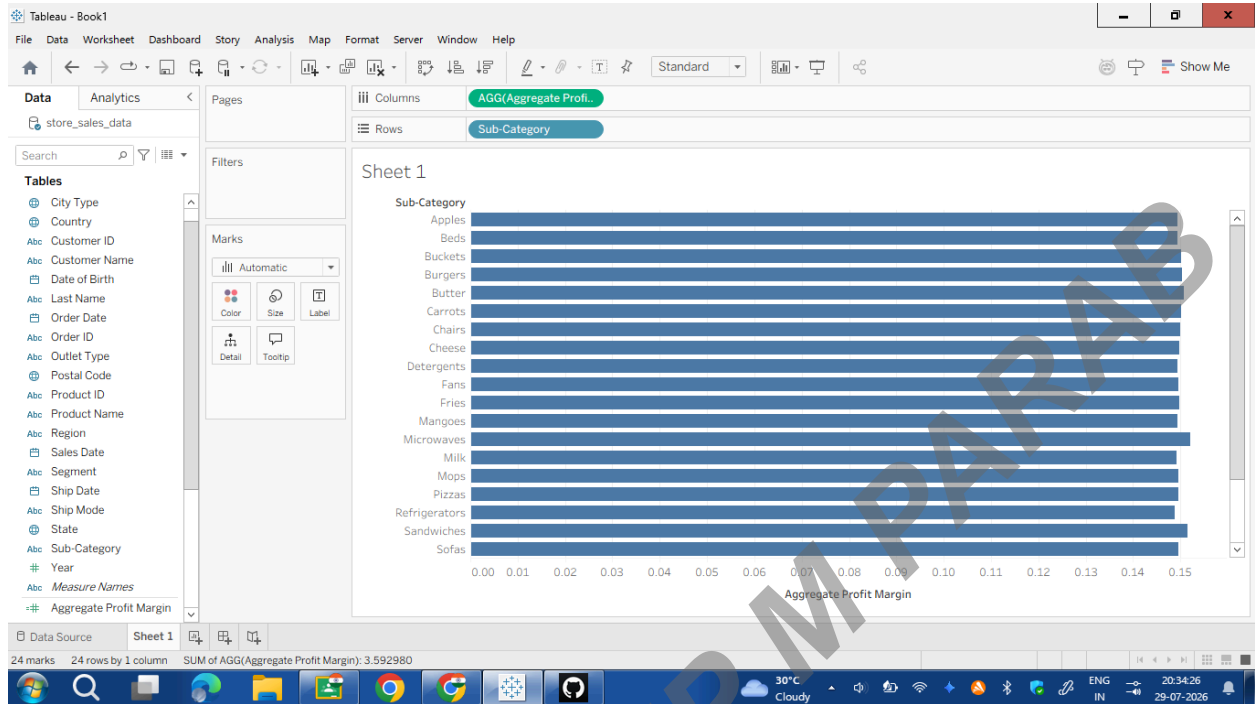


2. Putting the Field to Use

1. Drag the Sub-Category dimension from the Data Pane and drop it onto the Rows shelf.
2. Drag your newly created Aggregate Profit Margin calculated field and drop it onto the Columns shelf.
3. Observe closely: Look at the pill on the Columns shelf. It displays as AGG(Aggregate Profit Margin). Unlike standard fields, you cannot right-click this pill to change it to "Average" or "Count"—Tableau locks it as an aggregation because the math is already defined inside the formula.

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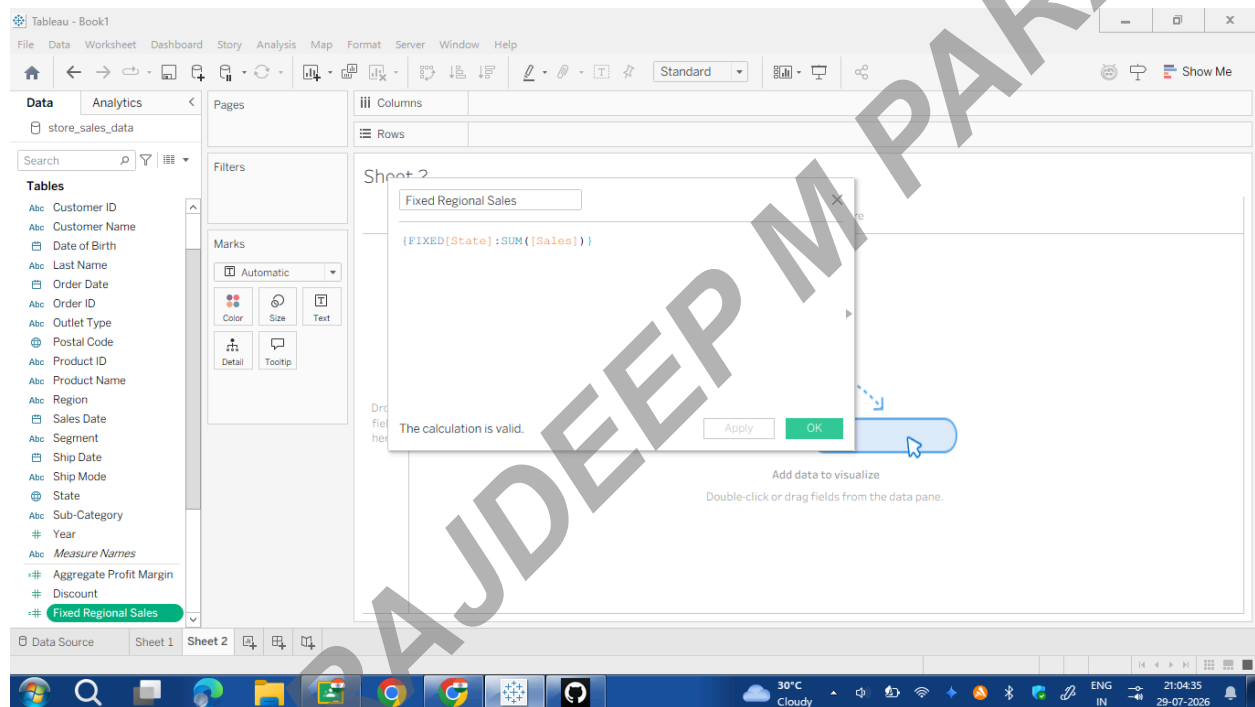
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D: Implementing FIXED Level of Detail (LOD) Calculations

1. Creating a FIXED LOD Calculated Field

1. Right-click in the blank white space of your Data Pane or Click the Small Triangle Icon next to search bar and select Create Calculated Field...
2. Name the field: Fixed Regional Sales.
3. In the calculation formula window, type the following expression using curly braces{}: { FIXED [State] : SUM([Sales]) }
4. Look at the bottom left to ensure it reads "The calculation is valid" and click OK.

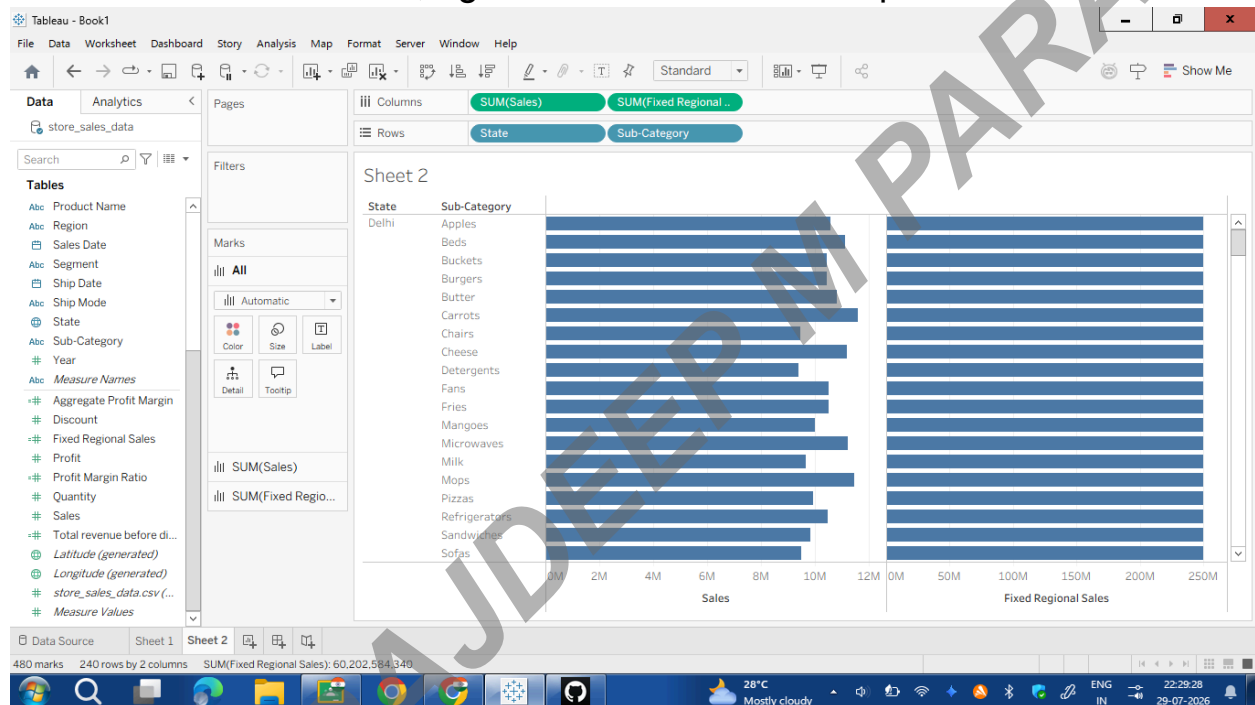


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2. Putting the Field to Use

1. Drag the State dimension from the Data Pane and drop it onto the Rows shelf.
2. Drag the Sub-Category dimension and drop it onto the Rows shelf, right next to State.
3. Drag the raw Sales measure pill onto the Columns shelf. (This shows the sales split down by each sub-category within that state).
4. Drag your newly created Fixed Regional Sales calculated field and drop it onto the Columns shelf, right next to the raw Sales pill.



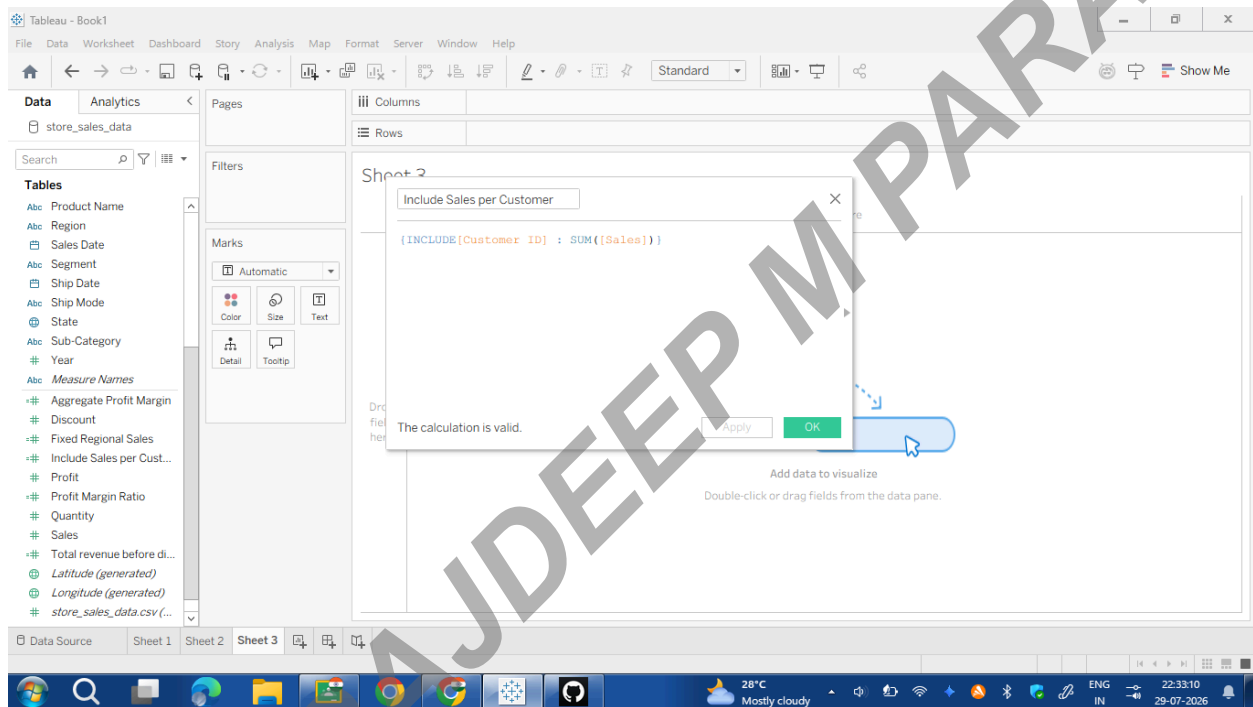
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E: Using INCLUDE and EXCLUDE LOD Expressions

1. Creating an INCLUDE LOD Calculated Field

1. Right-click in the blank white space of your Data Pane and select Create Calculated Field...
2. Name the field: Include Sales per Customer.
3. In the calculation formula canvas area, input the following expression:
{ INCLUDE [Customer ID] : SUM([Sales]) }
4. Click OK.

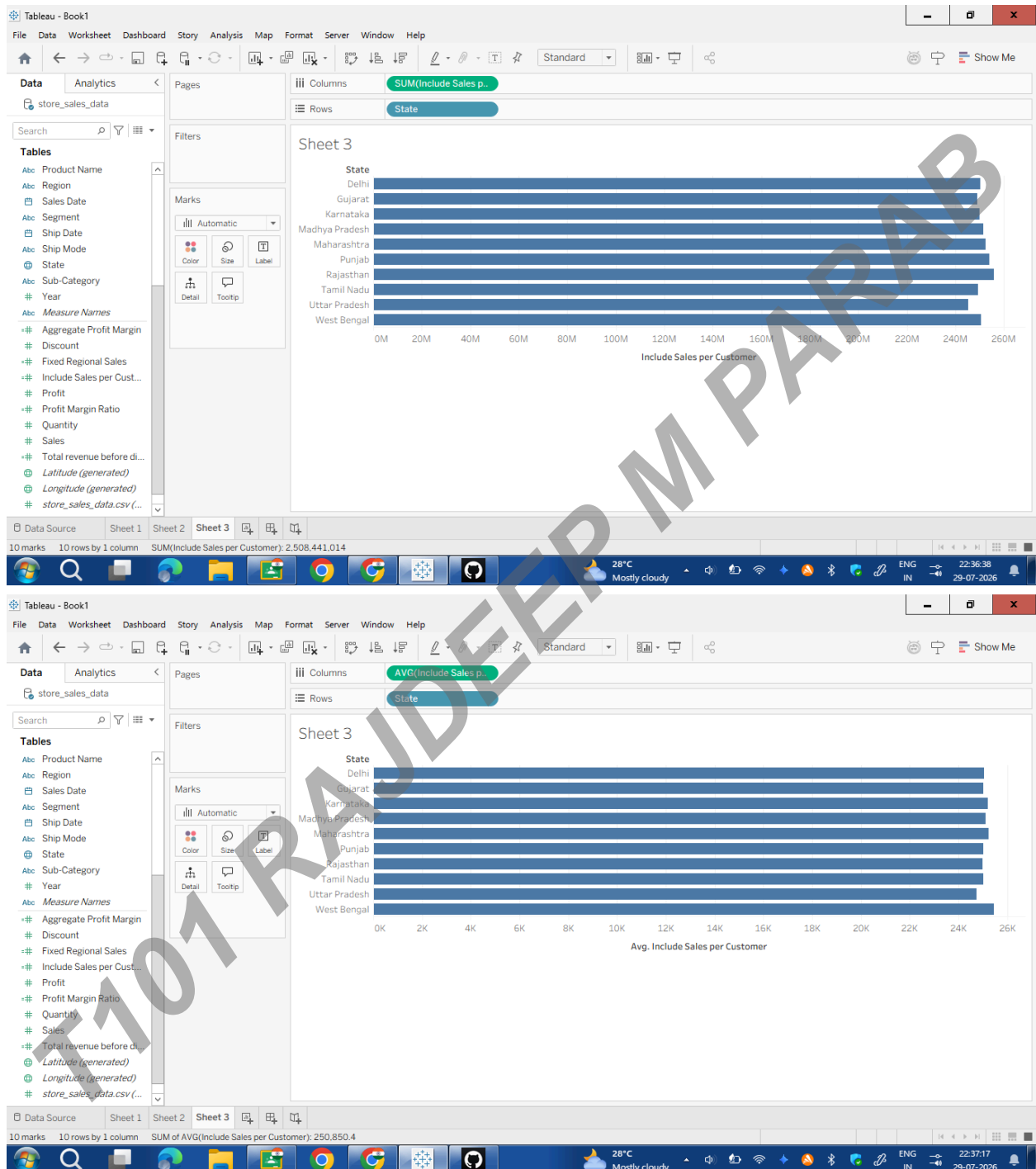


2. Putting the INCLUDE Field to Use

1. Drag the State dimension from the Data Pane and drop it onto the Rows shelf.
2. Drag your newly created Include Sales per Customer calculated field and drop it onto the Columns shelf.
3. Right-click the Include Sales per Customer pill on the Columns shelf, hover over Measure (Sum), and change it to Average.

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3. Creating an EXCLUDE LOD Calculated Field

1. Open a new blank worksheet.
2. Drag Category of Goods onto the Rows shelf.
3. Drag Sub-Category onto the Rows shelf (right next to Category of Goods).
4. Right-click in the blank white space of the Data Pane or Click the Small Triangle Icon next to search bar and select Create Calculated Field...
5. Name the field: Exclude Category Sales.
6. Input the following expression: { EXCLUDE [Sub-Category] :
SUM([Sales]) }
7. Click OK and drag this new field onto the Columns shelf.

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The top screenshot shows the Tableau interface with the 'store_sales_data' data source. The Columns shelf contains 'Category of Goods' and 'Sub-Category'. The Rows shelf is empty. The Marks shelf is set to 'Automatic'. The data table shows the following rows:

Category of Goods	Sub-Category	Sales
Dairy Products	Butter	Abc
Dairy Products	Cheese	Abc
Dairy Products	Milk	Abc
Dairy Products	Yogurt	Abc
Electric Appliances	Fans	Abc
Electric Appliances	Microwaves	Abc
Electric Appliances	Refrigerators	Abc
Electric Appliances	Washing Machines	Abc
Fast Food	Burgers	Abc
Fast Food	Fries	Abc
Fast Food	Pizzas	Abc
Fast Food	Sandwiches	Abc
Furniture	Beds	Abc
Furniture	Chairs	Abc
Furniture	Sofas	Abc
Furniture	Tables	Abc
Household Items	Buckets	Abc
Household Items	Detergents	Abc
Household Items	Mops	Abc
Household Items	Utensils	Abc
Sessional Fruits &	Apples	Abc
Sessional Fruits &	Carrots	Abc

The bottom screenshot shows the same interface with a calculated field 'Exclude category Sales' added to the data source. The calculated field is defined as:

```
[EXCLUDE [Sub-Category]] : SUM([Sales])
```

The calculation is valid. The data table shows the following rows:

Category of Goods	Sub-Category	Sales
Dairy Products	Butter	Abc
Dairy Products	Cheese	Abc
Dairy Products	Milk	Abc
Dairy Products	Yogurt	Abc
Electric Appliances	Fans	Abc
Electric Appliances	Microwaves	Abc
Electric Appliances	Refrigerators	Abc
Electric Appliances	Washing Machines	Abc
Fast Food	Burgers	Abc
Fast Food	Fries	Abc
Fast Food	Pizzas	Abc
Fast Food	Sandwiches	Abc
Furniture	Beds	Abc
Furniture	Chairs	Abc
Furniture	Sofas	Abc
Furniture	Tables	Abc
Household Items	Buckets	Abc
Household Items	Detergents	Abc
Household Items	Mops	Abc
Household Items	Utensils	Abc
Sessional Fruits &	Apples	Abc
Sessional Fruits &	Carrots	Abc

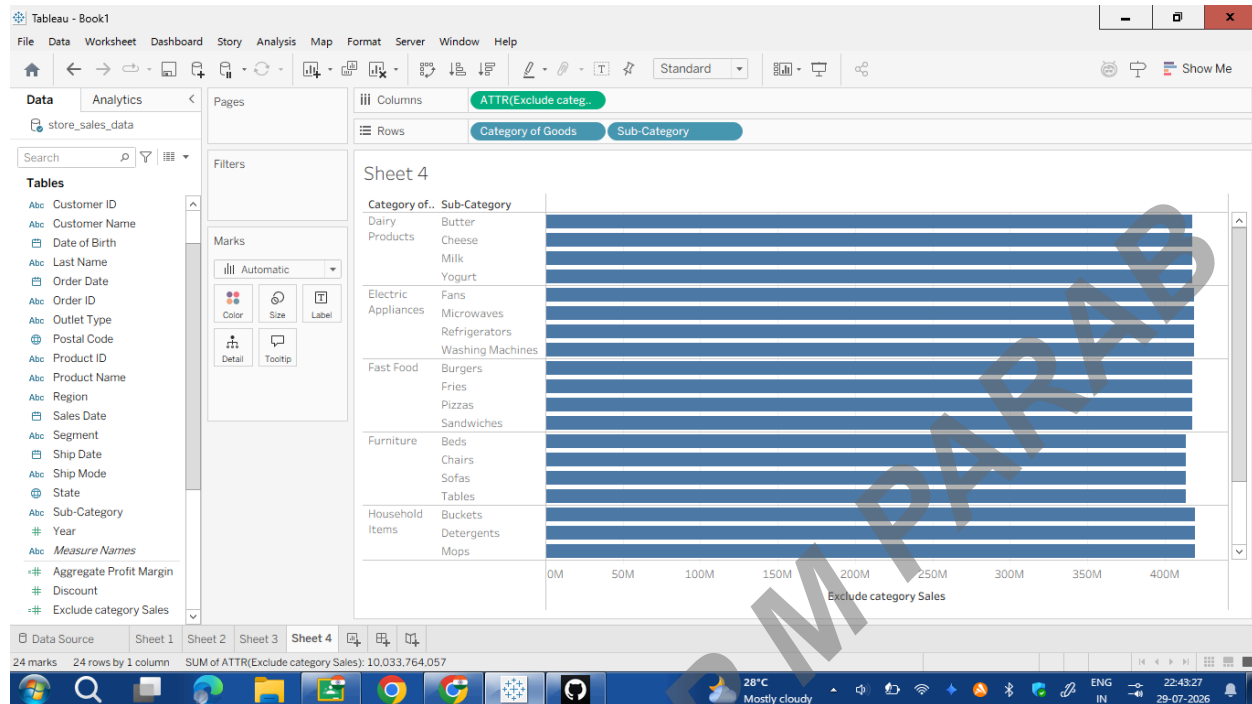
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F: Creating and Using Parameters for Dynamic Analysis

1. Creating a Parameter

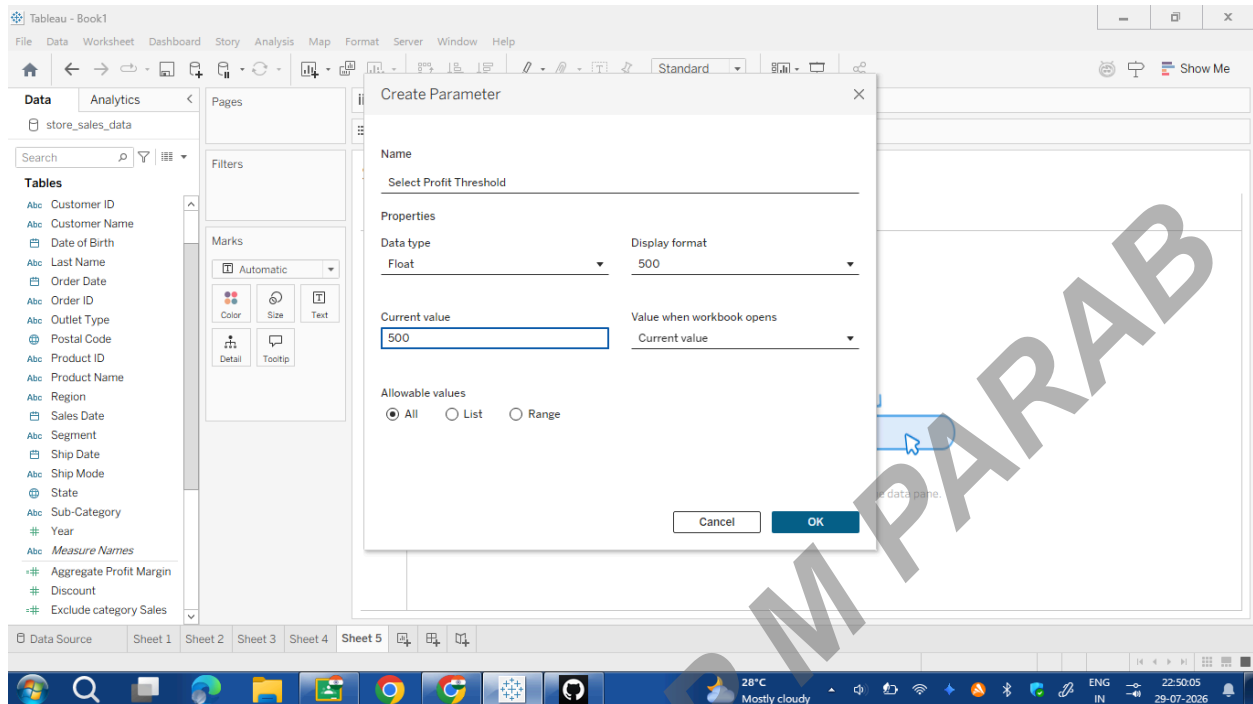
Unlike calculated fields which evaluate data statically, a Parameter is a dynamic variable or workbook input control block (like a slider or dropdown) that allows users to alter calculations or values on the fly without changing the underlying architecture.

Let's build a parameter that lets users set their own dynamic Profit Target threshold:

1. Go to your Data Pane. Click the small drop-down arrow (small triangle) next to the search bar at the top and choose Create Parameter.
2. Name the parameter: Select Profit Threshold.
3. Set the Data Type to Float (or Integer).
4. Set the Current Value to 500.
5. Keep the Allowable values set to All and click OK.
6. Notice a new section titled Parameters appears at the bottom of your Data Pane, showcasing your new selector.

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2. Linking the Parameter to a Calculated Field

A parameter does absolutely nothing until you actively tie it to a field layout or formula. Let's create a conditional formula that flags categories based on the parameter's current value:

1. Right-click in the blank white space of the Data Pane or Click on small triangle angle next to search bar and select Create Calculated Field...

2. Name this field: KPI Threshold Met?

3. Input the following logical IF-THEN conditional statement:

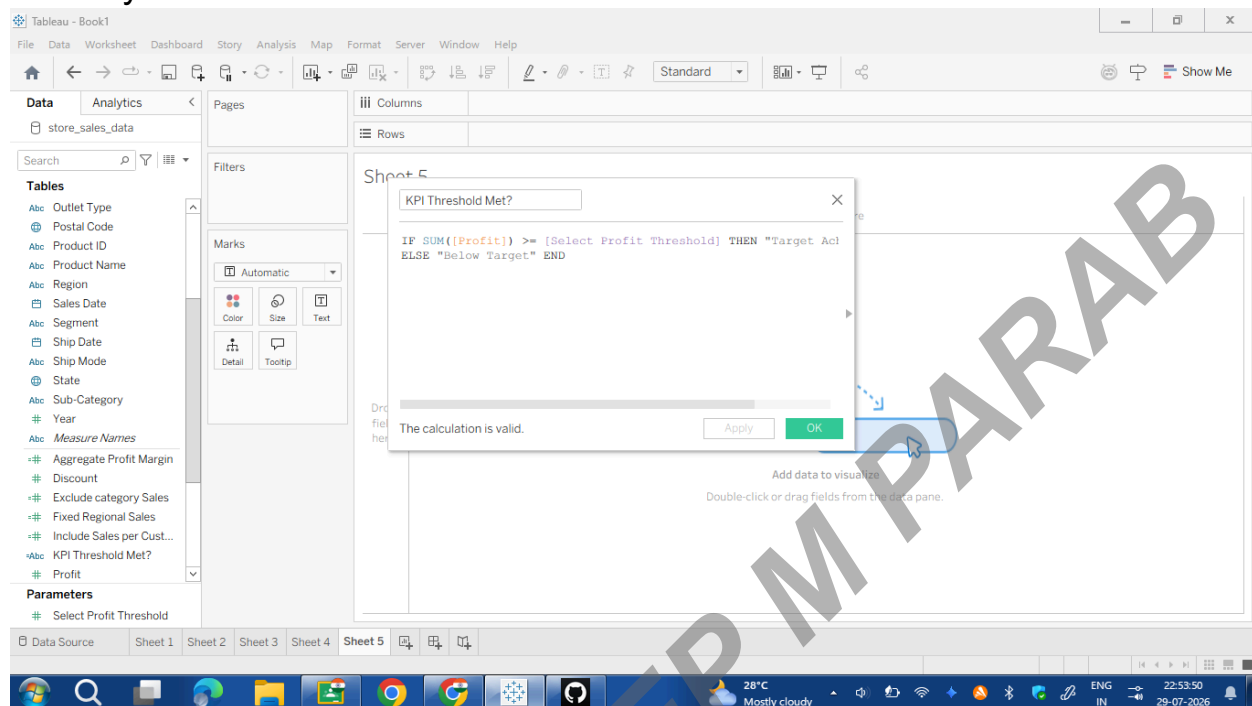
IF SUM([Profit]) >= [Select Profit Threshold] THEN "Target Achieved"
ELSE "Below Target" END

(NOTE: THE SELECT PROFIT THRESHOLD SHOULD TURN INTO PURPLE IF IT IS NOT TURNING IN PURPLE PLEASE RETYPE IT AGAIN)

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4. Verify the formula is valid and click OK.

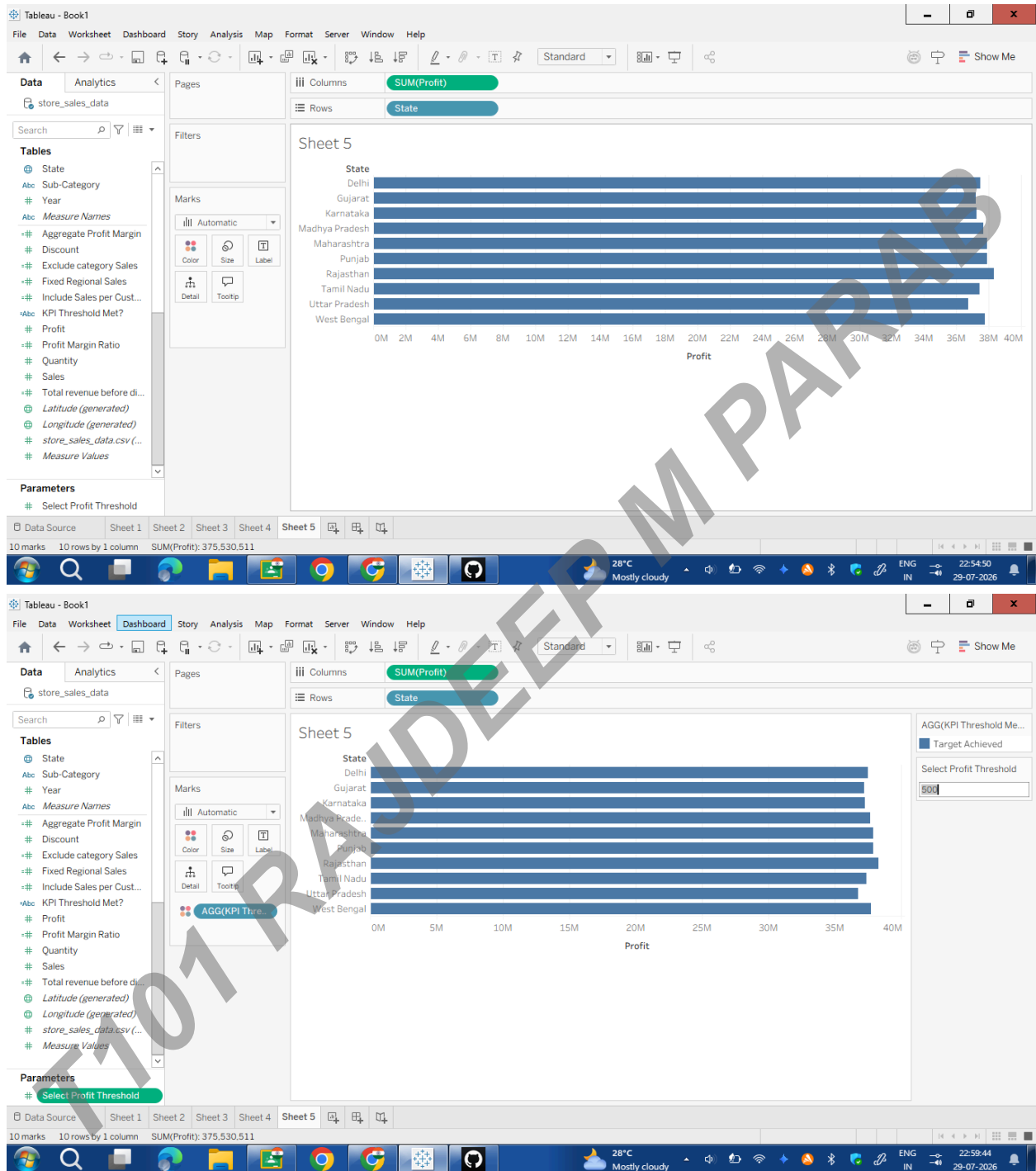


3. Putting the Interactive Parameter to Use

1. Open a new blank worksheet.
2. Drag the State dimension from your Data Pane and drop it onto the Rows shelf.
3. Drag the Profit measure and drop it onto the Columns shelf.
4. Drag your newly created calculated field KPI Threshold Met? and drop it directly onto the Color tile in the Marks card.
5. Show the Parameter Control: Go down to the very bottom of the Data Pane under the Parameters header section. Right-click on your Select Profit Threshold parameter block and choose Show Parameter Control.

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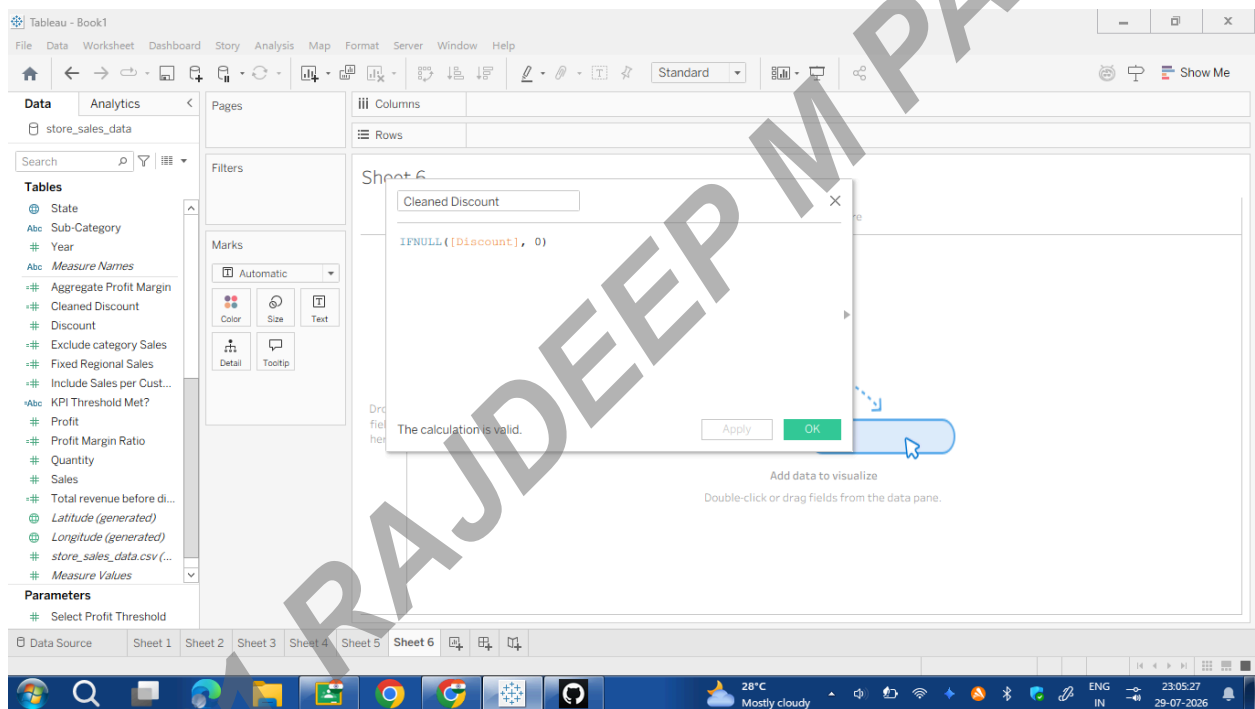
G: Apply Calculations to Fix Data Issues and Enhance Insights

1. Handling Null or Missing Values with IFNULL

In enterprise datasets, missing data or null entries can cause calculations to break or display empty spaces in charts. We can use a logical calculation to substitute a clean value whenever a null occurs.

Let's clean up potential missing indicators in our data:

1. Right-click in the blank white space of your Data Pane and select Create Calculated Field...
2. Name the field: Cleaned Discount.
3. Input the following expression: `IFNULL([Discount], 0)`
4. Click OK.



2. Segmenting Data for Insights using String Functions (UPPER/LOWER)

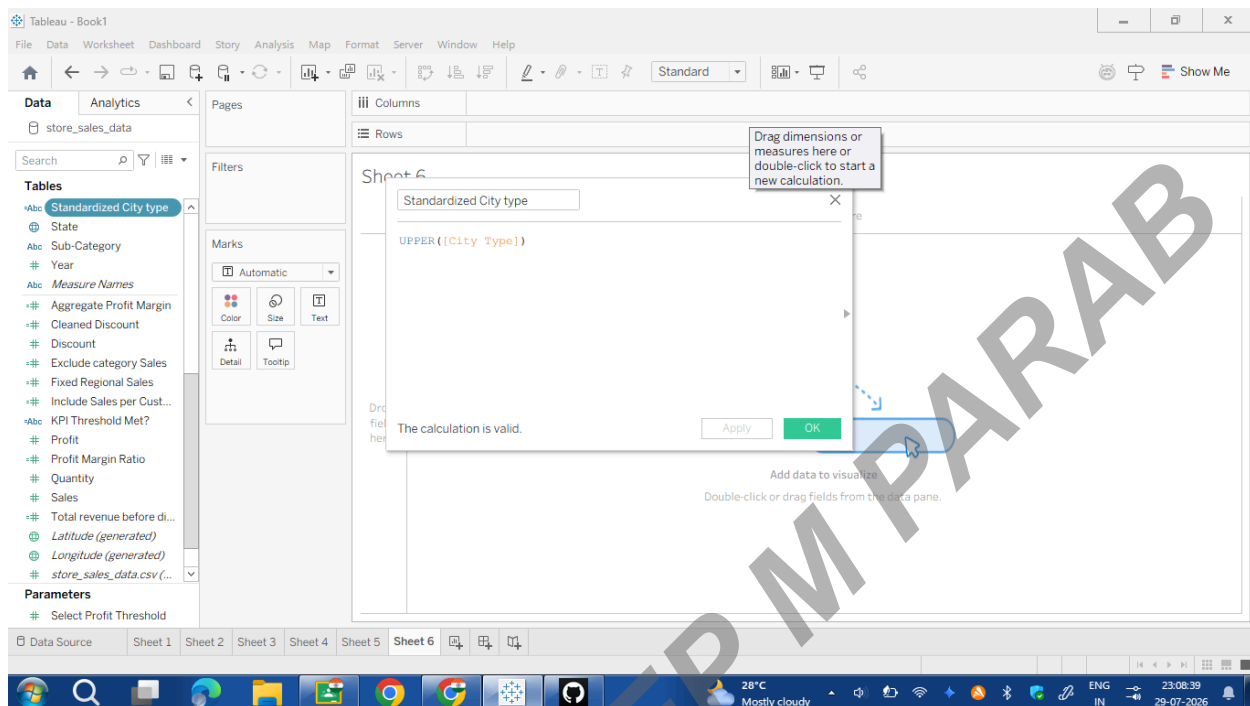
Sometimes database values are entered with inconsistent casing (e.g., "tier 1", "Tier 1", "TIER 1"), which causes Tableau to split them into separate visual blocks. We can force uniform compliance using a string calculation:

1. Create a new Calculated Field named Standardised City Type.
2. Input the following text transformation expression: `UPPER([City Type])`

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3. Click OK.

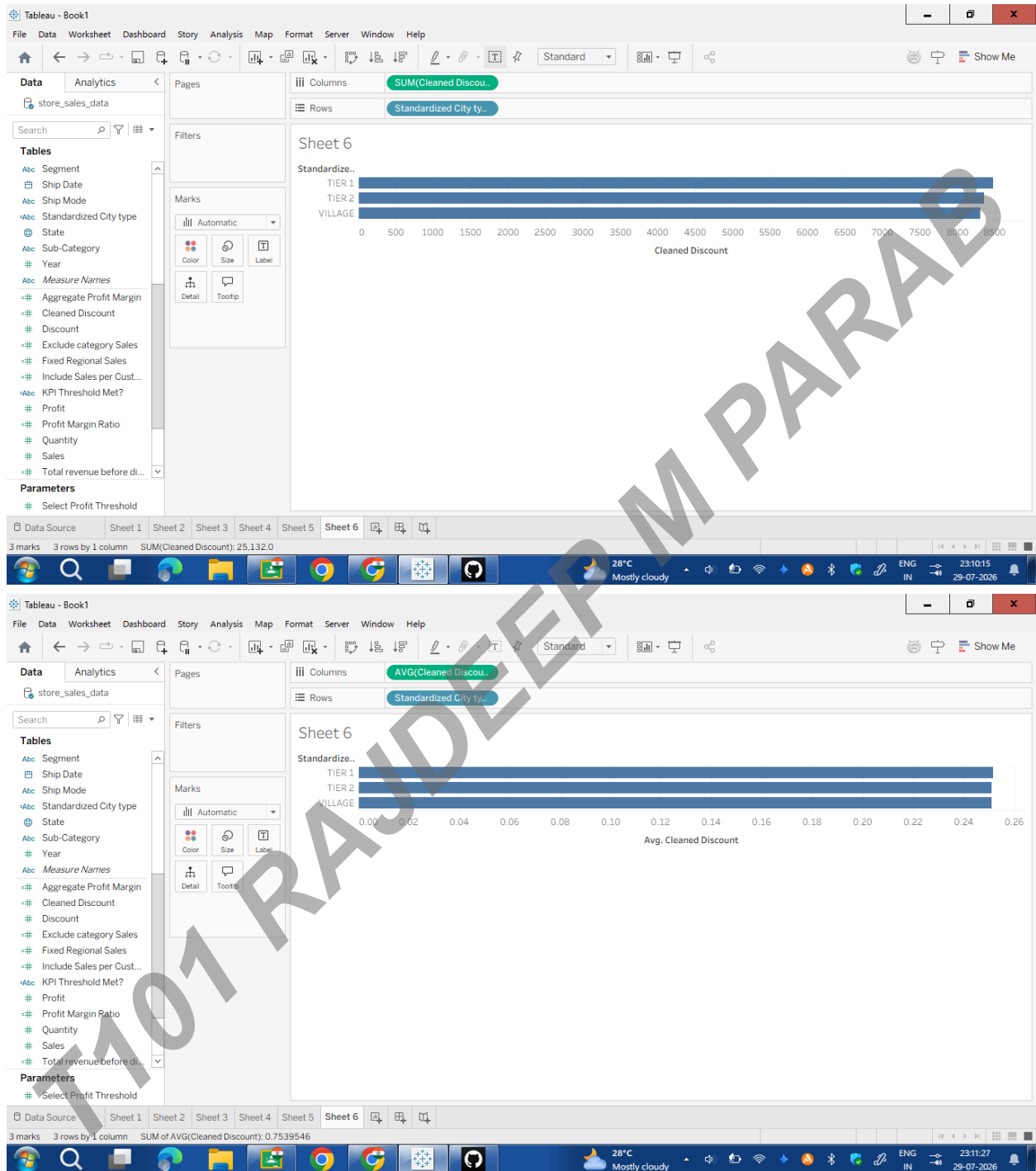


3. Putting the Cleaned Fields to Use

1. Open a new blank worksheet.
2. Drag your newly created Standardised City Type dimension from the Data Pane and drop it onto the Rows shelf.
3. Drag the Cleaned Discount calculated field onto the Columns shelf. Change its aggregation from Sum to Average by right-clicking the pill.

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